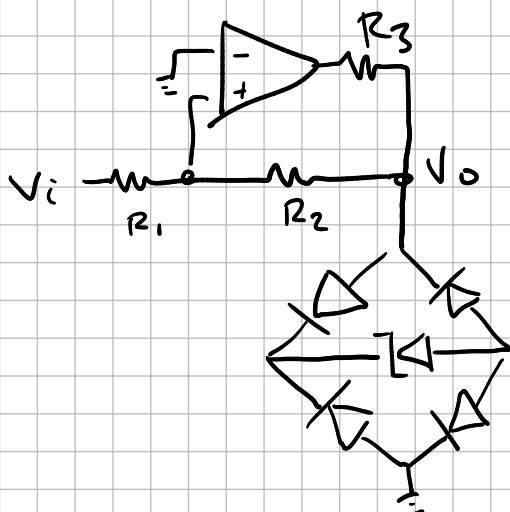


9.18]

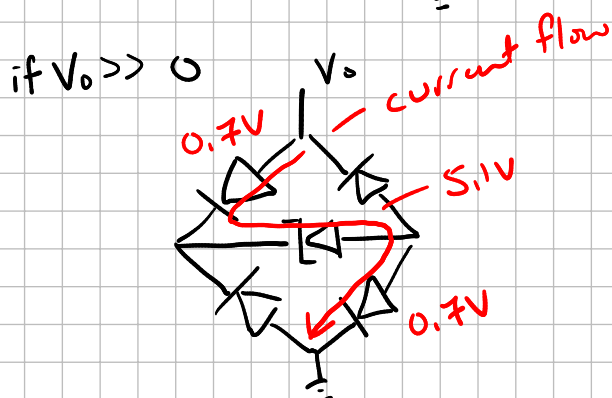


$$R_1 = 10k, R_2 = 13k, R_3 = 4.7k$$

$$V_Z = 5.1V, V_D = 0.7V$$

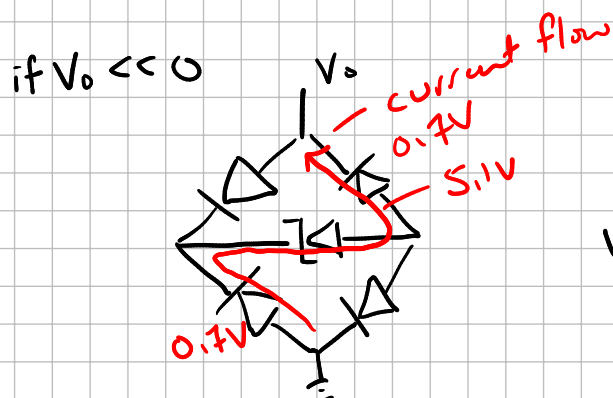
find VTC

Assume output of comparator is greater than or less than diode drops.



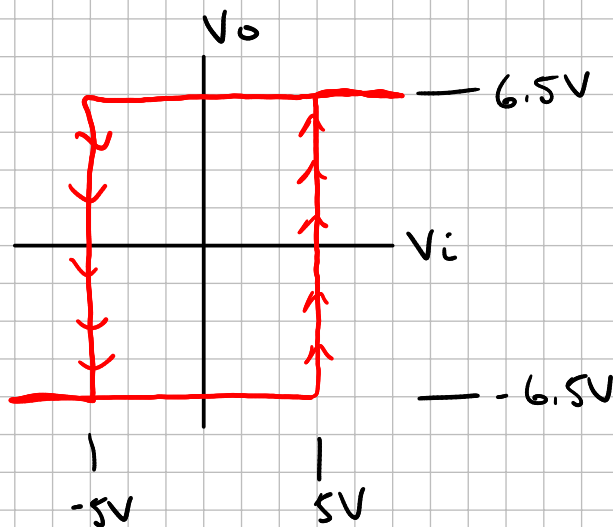
$$V_0 = 0 + 0.7V + 5.1V + 0.7V = 6.5V$$

$$V_i = V_{TH1} = -\frac{R_1}{R_2} V_0 = -\frac{10}{13} 6.5V = -5V$$

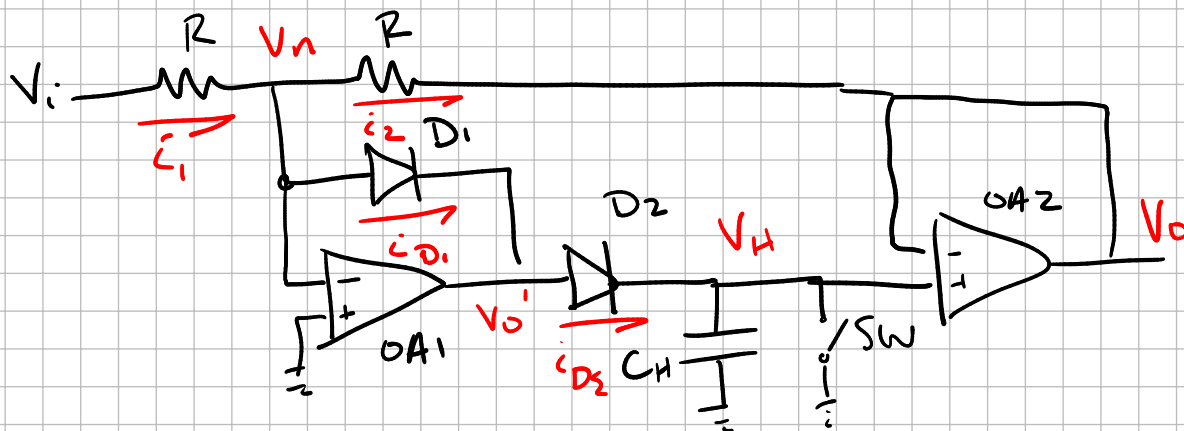


$$V_0 = 0 - 0.7V - 5.1V - 0.7V = -6.5V$$

$$V_i = V_{TH2} = -\frac{R_1}{R_2} V_0 = -\frac{10}{13} (-6.5V) = 5V$$



9.36 May be analyzed in LTspice with just a discussion from the student.
This solution does not use LTspice.



OA2 is a voltage follower, so $V_O = V_H$

Assuming virtual short is valid: $V_n = 0$

Start with no current through D1:

$V_i < 0$: i_1 is negative: flowing from $V_n \rightarrow V_i$

D1 must be off.

$$i_1 = i_2 \rightarrow \frac{V_i - V_n}{R} = \frac{V_n - V_O}{R} \rightarrow V_O = -V_i \quad (V_O \text{ is } > 0)$$

Since $V_O = V_H$, then $V_H = -V_i$. Since a capacitor

needs to charge, if $V_O' = -V_i + V_{O2}$ until $V_H = -V_i$

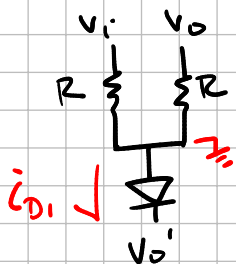
$V_i > 0$: we know the capacitor cannot discharge,

so V_O is fixed at some value, then we must have:

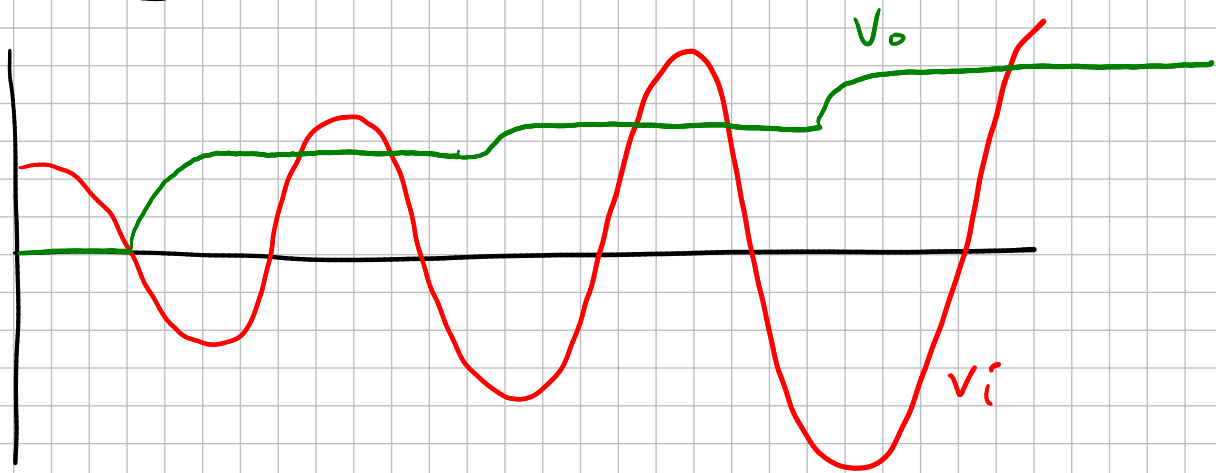
V_O' must be $V_O' = -V_{O1}$

$$\text{and } i_{D1} = \frac{V_i}{R} + \frac{V_O}{R} = (V_i + V_O)/R$$

D2 is off since $V_O' < 0V \nless V_H \geq 0V$

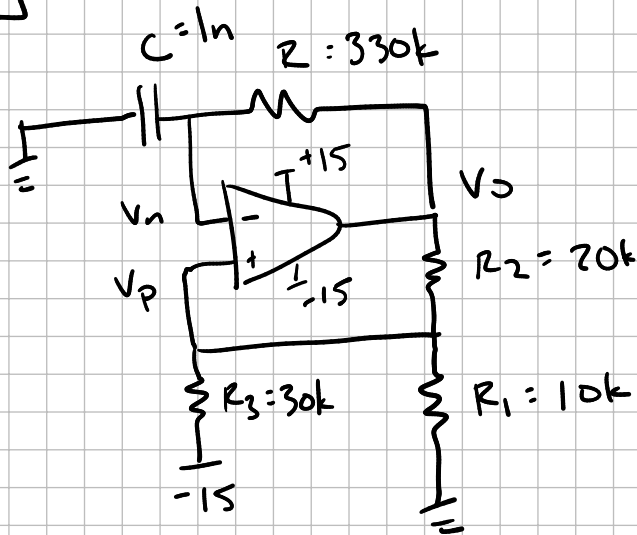


9.36 cont.



Assuming the capacitor charges quickly, the output will track the magnitude of the negative peaks of the input with a gain of: $-1V/V$
The gain would change with the resistor ratio: $-R_f/R_i$

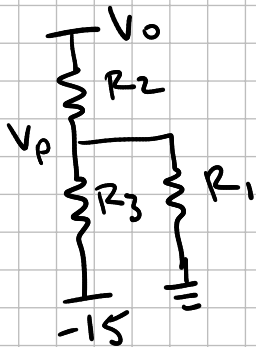
10.9



V_{OL}/V_{OH} not specified, let's assume $\pm 15V$ (rail-to-rail)

Note: text above fig 10.7 says $\pm 13V$ V_{OL}/V_{OH} , but that doesn't specify if $\pm 15V$ supplies are used.

Calculate V_p for $V_o = \pm 15V$:



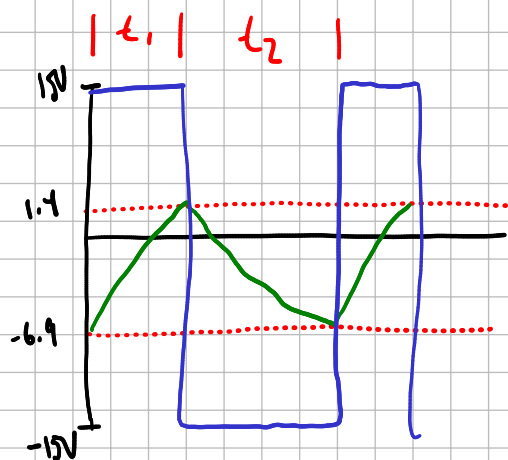
$$\frac{V_o - V_p}{R_2} = \frac{V_p - (-15)}{R_3} + \frac{V_p}{R_1}$$

$$\frac{V_o}{20} - \frac{15}{30} = V_p \left(\frac{1}{20} + \frac{1}{30} + \frac{1}{10} \right) = 0.18\bar{3} V_p$$

$$V_p = 0.27 V_o - 2.72$$

$$V_p = \begin{cases} -6.9V, & V_o = -15 \\ 1.4V, & V_o = 15 \end{cases} = V_{TH,1/2}$$

Cont $\longrightarrow$



$$R = 330k$$

$$C = 1n$$

find t_1 : $V_n = -6.9 + (15V - -6.9)(1 - e^{-t_1/RC}) = 1.4V$

$$t_1 = 157\mu s$$

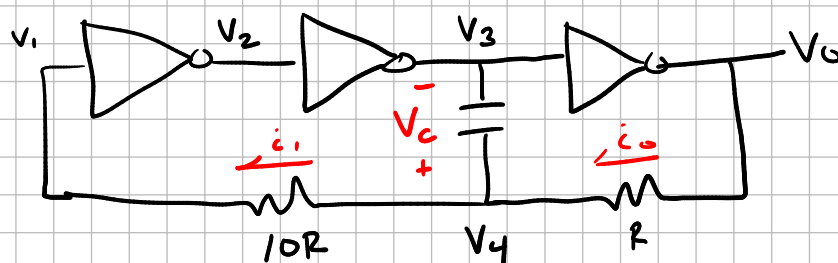
find t_2 : $V_n = 1.4 + (15V - 1.4)(1 - e^{-t_2/RC}) = -6.9V$

$$t_2 = 233\mu s$$

$$T = t_1 + t_2 = 390\mu s \rightarrow f = 2.56kHz$$

$$DC = 100 \times \frac{t_1}{T} = 40.3\%$$

10.14



$$V_T = 0.5V_{DD}$$

$$i_1 = 0 \rightarrow V_1 = V_4$$

$$i_0 = \frac{V_0 - V_4}{R}$$

$$V_1 < 0.5V_{DD}, V_2 = V_{DD}, V_3 = 0, V_0 = V_{DD}$$

$\rightarrow V_4$ charges towards V_{DD} until $V_4 = 0.5V_{DD}$
 $V_c = 0.5V_{DD}$

$$V_1 > 0.5V_{DD}, V_2 = 0, V_3 = V_{DD}, V_0 = 0$$

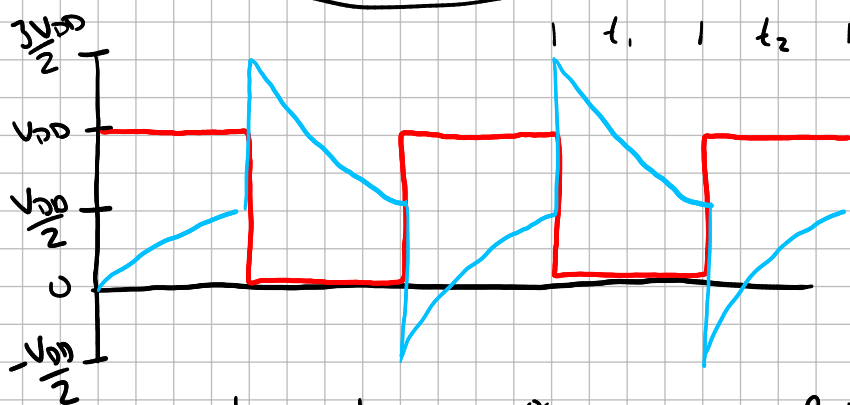
Initially, $V_c = 0.5V_{DD}$ & $V_3 = V_{DD}$, therefore, $V_4 = 1.5V_{DD}$

V_4 discharges towards 0 until $V_4 = 0.5V_{DD}$

$$V_c = -0.5V_{DD}$$

$$V_1 < 0.5V_{DD}$$

Initially, $V_c = 0.5V_{DD}$ & $V_3 = 0$, $\therefore V_4 = -0.5V_{DD}$



Duty cycle is 50%, $t_1 = t_2$. Find t_1 (or t_2)

$$\text{charge to } \frac{V_{DD}}{2}: \quad \frac{V_{DD}}{2} = -\frac{V_{DD}}{2} + \left(V_{DD} - \frac{V_{DD}}{2}\right)(1 - e^{-t_1/RC})$$

$$V_{DD} = \frac{3}{2}V_{DD}(1 - e^{-t_1/RC}) \rightarrow \frac{2}{3} = 1 - e^{-t_1/RC}$$

$$e^{-t_1/RC} = \frac{1}{3} \quad -t_1/RC = \ln\left(\frac{1}{3}\right) = -1.1, \quad t_1 = 1.1RC$$

$$T = 2.2RC, \quad f = \frac{1}{2.2RC}$$